

~/Desktop/Work/Stata/Do_Files/boston_replication_do_file.do

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1 *****
2 * PROJECT: Replication of 1996 Boston Fed Racial Disparities Study
3 * DATA: 2023 HMDA Public Extract
4 * AUTHOR: Benjamin D'Alessandro
5 * SOFTWARE: Stata 19.0
6 * DATE: March 2026
7 *****
8
9 ** Environment Setup **
10 version 19.0
11 clear all
12 macro drop _all
13 set more off
14 capture log close
15 cd "/Users/bendalessandro/Desktop/Work/Stata"
16 foreach pkg in estout {
17     capture which `pkg'
18     if _rc ssc install `pkg'
19 }
20 capture mkdir "Outputs"
21 log using "Outputs/boston_replication_log.txt", replace
22 import delimited "Raw_Data/boston_2023_hmda_subset.csv", clear
23
24 ** Variable Cleaning **
25
26 * 1. Initial Filtering *
27 replace debt_to_income_ratio = "" if inlist(debt_to_income_ratio, "Exempt", "NA")
28 keep if !missing(debt_to_income_ratio)
29 keep if loan_purpose == 1
30 keep if occupancy_type == 1
31 keep if derived_dwelling_category == "Single Family (1-4 Units):Site-Built"
32 keep if loan_type == 1
33 keep if lien_status == 1
34 keep if openend_line_of_credit == 2
35 keep if reverse_mortgage == 2
36 keep if business_or_commercial_purpose == 2
37 drop if inlist(derived_race, "Race Not Available", "Free Form Text Only", ///
38     "Joint")
39 drop if inlist(derived_ethnicity, "Ethnicity Not Available", "Free Form Text
40     Only", ///
41     "Joint")

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42 * 2. Destrting Variables *
43 foreach var in loan_to_value_ratio property_value income loan_amount {
44     destrting `var', replace ignore("Exempt" "NA" " ") force
45 }
46
47 encode lei, gen(lei_long)
48
49 * 3. Outlier Removal (Non-Positive or Top 1%) *
50 drop if income <= 0 | income == .
51 drop if loan_amount == .
52 drop if property_value == .
53 drop if loan_to_value_ratio == .
54
55 sum income, detail
56 local p99_income = r(p99)
57 sum loan_amount, detail
58 local p99_loan = r(p99)
59 sum property_value, detail
60 local p99_property = r(p99)
61
62 drop if income > `p99_income'
63 drop if loan_amount > `p99_loan'
64 drop if property_value > `p99_property'
65
66 * 4. Categorical and Natural Logarithm Variable Generation *
67 gen dti_cat = .
68 replace dti_cat = 1 if inlist(debt_to_income_ratio, "<20%", "20%<30%", ///
69     "30%<36%")
70 replace dti_cat = 2 if debt_to_income_ratio == "36%<43%"
71 replace dti_cat = 3 if debt_to_income_ratio == "43%<50%"
72 replace dti_cat = 4 if inlist(debt_to_income_ratio, "50%<60%", ">60%")
73
74 destrting debt_to_income_ratio, gen(dti_num) force
75 replace dti_cat = 1 if dti_num < 36 & missing(dti_cat)
76 replace dti_cat = 2 if dti_num >= 36 & dti_num < 43 & missing(dti_cat)
77 replace dti_cat = 3 if dti_num >= 43 & dti_num <= 50 & missing(dti_cat)
78 replace dti_cat = 4 if dti_num > 50 & missing(dti_cat)
79
80 label define dti_lbl 1 "<36%" 2 "36-42%" 3 "43-50%" 4 ">50%"
81 label values dti_cat dti_lbl
82
83 gen aus_reported = 0
84 replace aus_reported = 1 if inlist(aus1, 1, 2, 3, 4, 5, 7)
85
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86 gen aus_gse = 0
87 gen aus_internal = 0
88 gen aus_other = 0

89
90 foreach ausvar in aus1 aus2 aus3 aus4 aus5 {
91     replace aus_gse = 1 if inlist(`ausvar', 1, 2)
92     replace aus_internal = 1 if `ausvar' == 7
93     replace aus_other = 1 if inlist(`ausvar', 3, 4, 5)
94 }
95
96 gen ln_income = ln(income)
97
98 * 5. Outcome and Race/Ethnicity Definitions *
99 gen denied = (action_taken == 3) if inlist(action_taken, 1, 3)
100 label define denied_lbl 0 "Approved" 1 "Denied"
101 label values denied denied_lbl
102
103 gen white_nh = (derived_race == "White" & derived_ethnicity == "Not Hispanic or
    Latino")
104
105 gen minority = (white_nh == 0) if !missing(white_nh)
106 label var minority "Minority"
107
108 gen race_eth = derived_race
109 replace race_eth = "Black" if derived_race == "Black or African American"
110 replace race_eth = "Hispanic" if derived_race == "White" & ///
111     derived_ethnicity == "Hispanic or Latino"
112 replace race_eth = "Native Hawaiian or Pacific Islander" if ///
113     derived_race == "Native Hawaiian or Other Pacific Islander"
114 encode race_eth, gen(granular_race)
115
116 gen race_collapsed = race_eth
117 replace race_collapsed = "Non-Primary Minority" if inlist(derived_race, ///
118     "American Indian or Alaska Native", ///
119     "Native Hawaiian or Other Pacific Islander", ///
120     "2 or more minority races")
121 encode race_collapsed, gen(race)
122
123 ** Analysis & Visualizations **
124
125 * 1. Font Setup for Visualizations *
126 graph set window fontface "Times New Roman"
127 graph set print fontface "Times New Roman"
128

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129 * 2. Descriptive Statistics *
130 tabulate denied granular_race, row
131 tabulate granular_race
132
133 tabstat income, by(granular_race) statistics(median sd) columns(statistics)
134 tabstat dti_num loan_to_value_ratio, by(granular_race) statistics(mean sd) ///
135     columns(statistics)
136
137 * 3. Model 1: Denial Disparities for All Minority *
138 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat, vce(cluster
139     lei_long)
140 eststo m1
141 esttab m1 using "Outputs/reg_m1_minority.tex", ///
142     replace b(3) se(3) booktabs nomtitles nonumber label ///
143     title("Logit Results for Denial of Minorities") ///
144     addnote("Standard errors clustered by lender in parentheses") ///
145     substitute("%" "\%" "<" "\textless{}" ">" "\textgreater{}")
146
147 margins minority
148 marginsplot, recast(scatter) ///
149     plotopts(msymbol(0) msize(vsmall) mcolor(navy) lcolor(navy)) ///
150     ciopts(recast(rcap) lcolor(navy)) ///
151     title("Figure 1: Predicted White vs. Minority Loan Denial Probability", ///
152         size(medium)) ///
153     ytitle("Probability of Denial") xtitle("") ///
154     xlabel(0 "White" 1 "Minority") ///
155     xscale(range(-0.5 1.5)) ///
156     ylabel(0(0.02)0.1) ///
157     aspectratio(0.8) ///
158     graphregion(fcolor(white) lcolor(white)) ///
159     plotregion(margin(small)) ///
160     note("Error bars represent a 95% confidence interval." ///
161         "Controls: Income, LTV, DTI.", size(vsmall)) ///
162     name(minority_estimates, replace)
163
164 graph export "Outputs/denial_minority.png", replace
165
166 margins, dydx(minority) post
167 eststo margins_m1
168 esttab margins_m1 using "Outputs/margins_m1_minority.tex", ///
169     replace b(3) se(3) booktabs nomtitles nonumber label ///
170     title("Average Marginal Effects for All Minority Applicants") ///
171     addnote("Standard errors clustered by lender in parentheses") ///
172     substitute("%" "\%" "<" "\textless{}" ">" "\textgreater{}")
173
174 * 4. Model 2: Denial by Race *

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173 levelsof race if race_eth == "White", local(white_code)
174 if "`white_code'" == "" {
175     di as error "WARNING: 'White' not found in race_eth; base category not set."
176 }
177 else {
178     fvset base `white_code' race
179 }
180 logit denied i.race ln_income loan_to_value_ratio i.dti_cat, vce(cluster lei_long)
181 eststo m2
182 esttab m2 using "Outputs/reg_m2_race.tex", ///
183     replace b(3) se(3) booktabs nomtitles nonumber label ///
184     title("Logit Results for Denial by Race") ///
185     addnote("Standard errors clustered by lender in parentheses") ///
186     substitute("%" "\%" "<" "\textless{}" ">" "\textgreater{}")
187
188 margins race
189 marginsplot, recast(scatter) ///
190     plotopts(msymbol(0) msize(vsmall) mcolor(navy) lcolor(navy)) ///
191     ciopts(recast(rcap) lcolor(navy)) ///
192     title("Figure 2: Predicted Loan Denial Probability by Racial and Ethnic
193 Group", ///
194     size(medium)) ///
195     ytitle("Probability of Denial") xtitle("") ///
196     xlabel(, angle(45) labsize(small)) ///
197     xscale(range(0.7 5.3)) ///
198     ylabel(0(0.05)0.2) ///
199     note("Error bars represent a 95% confidence interval." ///
200         "Controls: Income, LTV, DTI.", size(vsmall)) ///
201     name(race_estimates, replace)
202
203 graph export "Outputs/denial_by_race.png", replace
204
205 margins, dydx(race) post
206 eststo margins_m2
207 esttab margins_m2 using "Outputs/margins_m2_race.tex", ///
208     replace b(3) se(3) booktabs nomtitles nonumber label ///
209     title("Average Marginal Effects of Race on Denial") ///
210     addnote("Standard errors clustered by lender in parentheses") ///
211     substitute("%" "\%" "<" "\textless{}" ">" "\textgreater{}")
212
213 * 5. Model 3: Denial Disparities Controlling for Lender Fixed Effects *
214 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat i.lei_long, ///
215     vce(cluster lei_long)
216 eststo m3
217 esttab m3 using "Outputs/reg_m3_lender_fe.tex", ///
218     replace b(3) se(3) booktabs nomtitles nonumber label ///

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217     drop(*.lei_long) ///
218     title("Logit Results Controlling for Lender Fixed Effects") ///
219     addnote("Standard errors clustered by lender in parentheses") ///
220     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
221
222 margins minority
223 margins, dydx(minority) post
224 eststo margins_m3
225 esttab margins_m3 using "Outputs/margins_m3_lender_fe.tex", ///
226     replace b(3) se(3) booktabs nomtitles nonumber label ///
227     title("Average Marginal Effects for Lender Fixed Effects") ///
228     addnote("Standard errors clustered by lender in parentheses") ///
229     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
230
231 * 6. Model 4: High Income Sample *
232 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat ///
233     if income > 150, vce(cluster lei_long)
234 eststo m4
235 esttab m4 using "Outputs/reg_m4_high_income.tex", ///
236     replace b(3) se(3) booktabs nomtitles nonumber label ///
237     title("Logit Results for High-Income Sample") ///
238     addnote("Standard errors clustered by lender in parentheses") ///
239     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
240
241 margins, dydx(minority) post
242 eststo margins_m4
243 esttab margins_m4 using "Outputs/margins_m4_high_income.tex", ///
244     replace b(3) se(3) booktabs nomtitles nonumber label ///
245     title("Average Marginal Effects for High-Income Sample") ///
246     addnote("Standard errors clustered by lender in parentheses") ///
247     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
248
249 * 7. Model 5: Tract Control *
250 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat ///
251     tract_to_msa_income_percentage, vce(cluster lei_long)
252 eststo m5
253 esttab m5 using "Outputs/reg_m5_tract.tex", ///
254     replace b(3) se(3) booktabs nomtitles nonumber label ///
255     title("Logit Results with Tract Control") ///
256     addnote("Standard errors clustered by lender in parentheses") ///
257     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
258
259 margins, dydx(minority) post
260 eststo margins_m5
261 esttab margins_m5 using "Outputs/margins_m5_tract.tex", ///

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262     replace b(3) se(3) booktabs nomtitles nonumber label ///
263     title("Average Marginal Effects with Tract Control") ///
264     addnote("Standard errors clustered by lender in parentheses") ///
265     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
266
267 * 8. Model 6: AUS Usage Reported *
268 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat ///
269     i.aus_reported, vce(cluster lei_long)
270
271 eststo m6
272 esttab m6 using "Outputs/reg_m6_aus_reported.tex", ///
273     replace b(3) se(3) booktabs nomtitles nonumber label ///
274     title("Logit Results Controlling for Reported AUS Usage") ///
275     addnote("Standard errors clustered by lender in parentheses") ///
276     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
277
278 margins, dydx(minority) post
279 eststo margins_m6
280 esttab margins_m6 using "Outputs/margins_m6_aus_reported.tex", ///
281     replace b(3) se(3) booktabs nomtitles nonumber label ///
282     title("Average Marginal Effects for Reported AUS Usage") ///
283     addnote("Standard errors clustered by lender in parentheses") ///
284     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
285
286 * 9. Model 7: AUS Type *
287 logit denied i.minority ln_income loan_to_value_ratio i.dti_cat ///
288     aus_gse aus_internal aus_other, vce(cluster lei_long)
289 eststo m7
290 esttab m7 using "Outputs/reg_m7_aus_type.tex", ///
291     replace b(3) se(3) booktabs nomtitles nonumber label ///
292     title("Logit Results Controlling for AUS Model") ///
293     addnote("Standard errors clustered by lender in parentheses") ///
294     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
295
296 margins, dydx(minority) post
297 eststo margins_m7
298 esttab margins_m7 using "Outputs/margins_m7_aus_type.tex", ///
299     replace b(3) se(3) booktabs nomtitles nonumber label ///
300     title("Average Marginal Effects for AUS Model") ///
301     addnote("Standard errors clustered by lender in parentheses") ///
302     substitute("%" "\%" "<" "\textless{" ">" "\textgreater{")
303
304 * 10. Model 8: Link Test *
305 quietly logit denied i.minority ln_income loan_to_value_ratio i.dti_cat, ///
    vce(cluster lei_long)

```

```
306 | linktest
307 |
308 | log close
```